



Generating Synthetic Health Data for Infectious Diseases

Benjamin Grime

School of Computing and Communications, Lancaster University



Background

Clinical trial data is difficult to share publicly due to privacy risks, such as participant re-identification. This research focuses on generating synthetic data for infectious diseases, particularly malaria, whilst maintaining both utility and privacy. Synthetic data can mimic real datasets while masking sensitive information, facilitating more efficient data sharing.

Aim - a synthetic dataset that satisfies analytical tests on utility and privacy.

- Using models based on real data, we can generate 'fake', synthetic data.
- The goal is to preserve the relationships and distributions present in the real data (*utility*).
- Must mask the original data such that it is difficult for an attacker to infer any real information present in the original dataset (*privacy*).

Generation Method

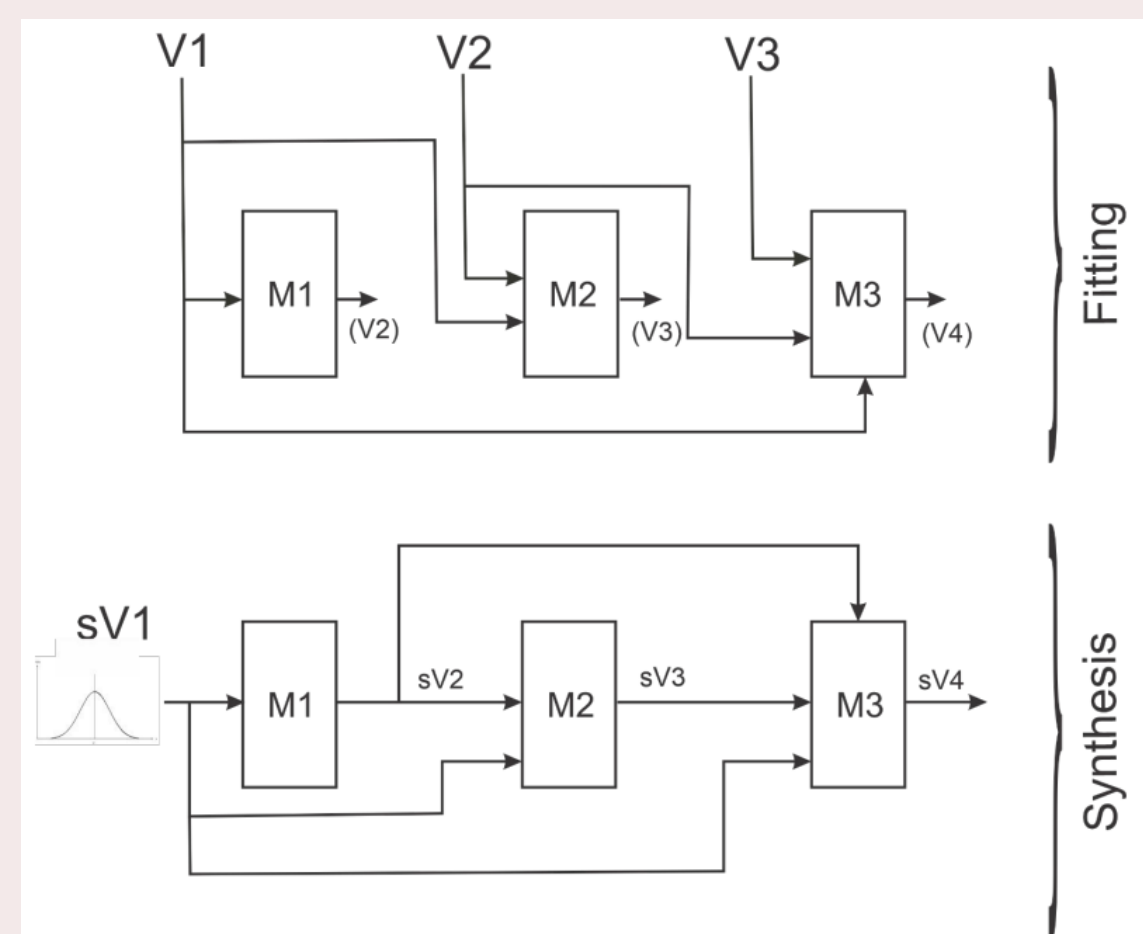


Figure 1: An illustration of the synthetic process using sequential regression modelling

Synthetic data of 3,428 malarial clinical trial subjects was generated using Classification and Regression Trees (CARTs) within the Sequential Regression Modeling (SRM) via the *synthpop* package in R.

- Fitting - CARTs were used to model both categorical and continuous variables. They were fitted to each real variable given all previous variables (i.e. all those to the left).
- Synthesis - initialised by sampling with replacement from the first variable real variable to produce the first synthetic variable. The next synthetic variables are generated iteratively by sampling from the predicted terminal node of the fitted CARTs conditional on the previously synthesized variables.

Evaluative Metrics

Utility:

- Graphical comparisons - qualitative / visual similarity of the marginal distributions of the real and synthetic data.
- Linear mixed models (LMMs) - an extension of linear models used to predict hemoglobin values over time. Includes random intercepts at the subject level. The key here is not to optimise for performance, but to see whether models trained on real or synthetic data perform similarly and equally capture the marginal distributions of longitudinal variables.
- Distinguishability - gradient boosted decision tree (GBDT) distinguishes between real and synthetic data, where $p_i = P(\text{real})$:

$$d = \frac{1}{N} \sum_i (p_i - 0.5)^2$$

Privacy:

- For 1,000 records, fully estimating privacy takes up to 250 days of compute - infeasible [1].
- Several variations of multiple metrics must be implemented to obtain a solid working estimate of privacy.
- Opted for synthetic data that was at least first utilisable and passed sufficient quality checks before thoroughly evaluating privacy - privacy could not be evaluated.

Results

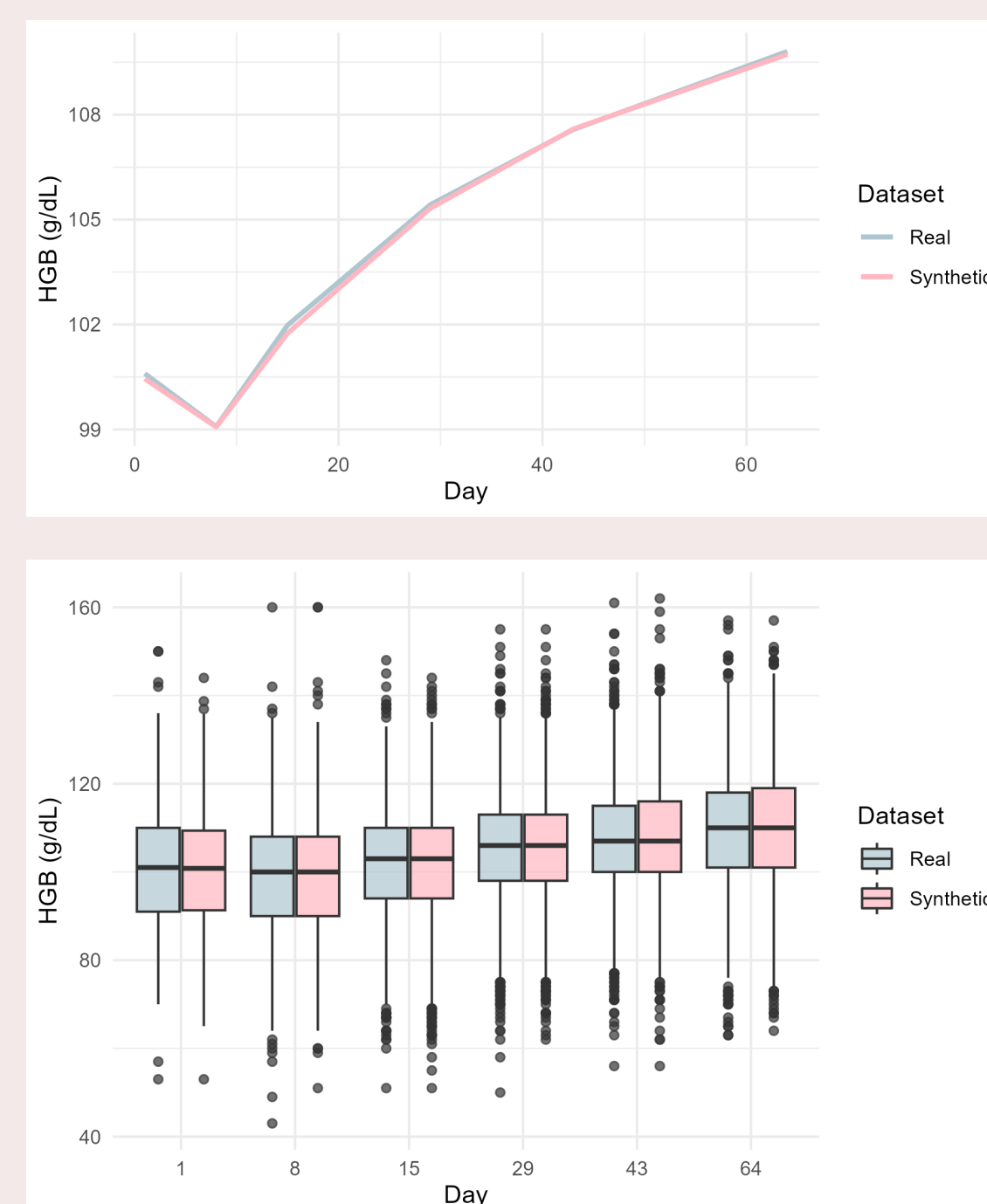


Figure 2: Marginal distributions of hemoglobin values over time

Graphs indicate continuous variables were synthesised well, with similar results for categorical variables. Outliers may have been replicated too closely, compromising privacy.

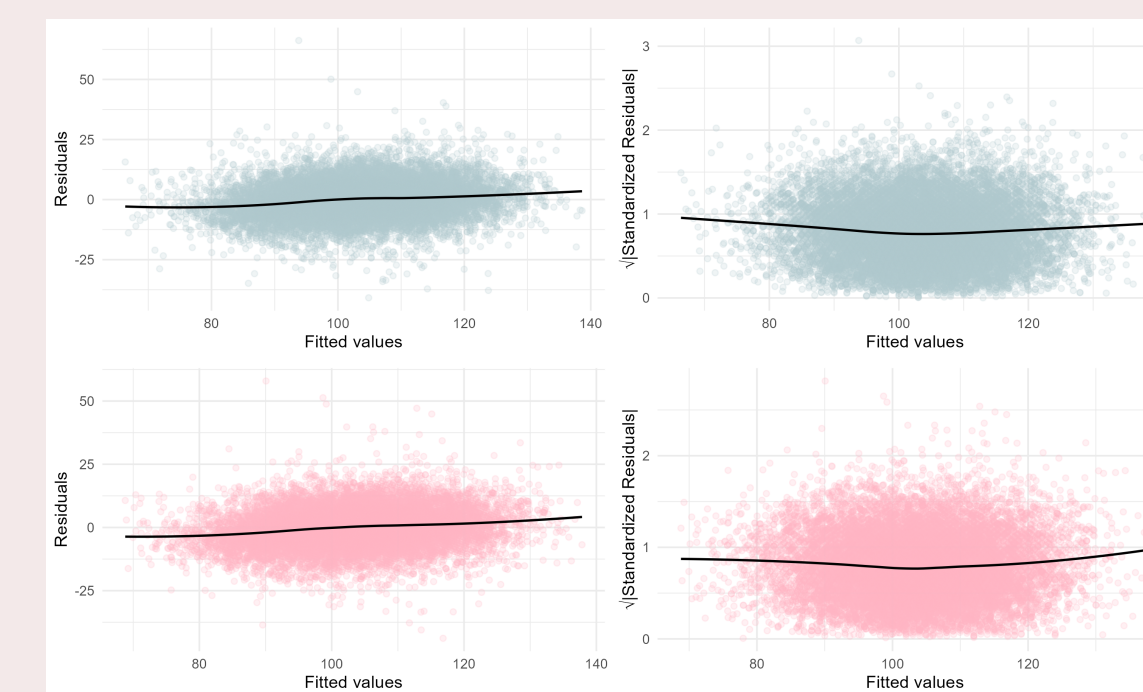


Figure 3: Residuals and scale-location plots of real and synthetic LMMs

$$\begin{aligned} \text{Real}_{ij} &= 99.6 + 0.168x_{ij} + b_i + \epsilon_{ij} \\ b_i &\sim \mathcal{N}(0, 101.6) \quad \epsilon_{ij} \sim \mathcal{N}(0, 60.7) \end{aligned}$$

$$\begin{aligned} \text{Syn}_{ij} &= 99.4 + 0.168x_{ij} + b_i + \epsilon_{ij} \\ b_i &\sim \mathcal{N}(0, 97.4) \quad \epsilon_{ij} \sim \mathcal{N}(0, 63.1) \end{aligned}$$

LMMs show strong similarities, suggesting utilisable synthetic data.

Tree complexity is positively correlated with distinguishability but inversely correlated with AUROC, where the AUROCs for all trees were below 0.5. This has an unintuitive yet well-established interpretation of random guessing [2]. Thus, the synthetic data hides well amongst the real data.

Conclusion

There is reasonable evidence that our synthetic data has good utility:

- Both continuous and categorical variables were replicated well.
- LMMs trained on real and synthetic data were similar.
- GBDTs failed to distinguish between real and synthetic data.

The current approach could be improved by:

- Evaluating privacy.
- Including more variables to assess similarity of conclusions regarding clinically relevant outcomes.

References

- [1] Matthieu Meeus and Florent Guépin (2024) Achilles' Heels: Vulnerable Record Identification in Synthetic Data Publishing
- [2] Jamalabadi, Alizadeh, Schönauer, Leibold, Gais (2016) In Human Brain Mapping Classification based hypothesis testing in neuroscience: Below-chance level classification rates and overlooked statistical properties of linear parametric classifiers